

Model Paper - Database Management Systems

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Q1 (20 marks)

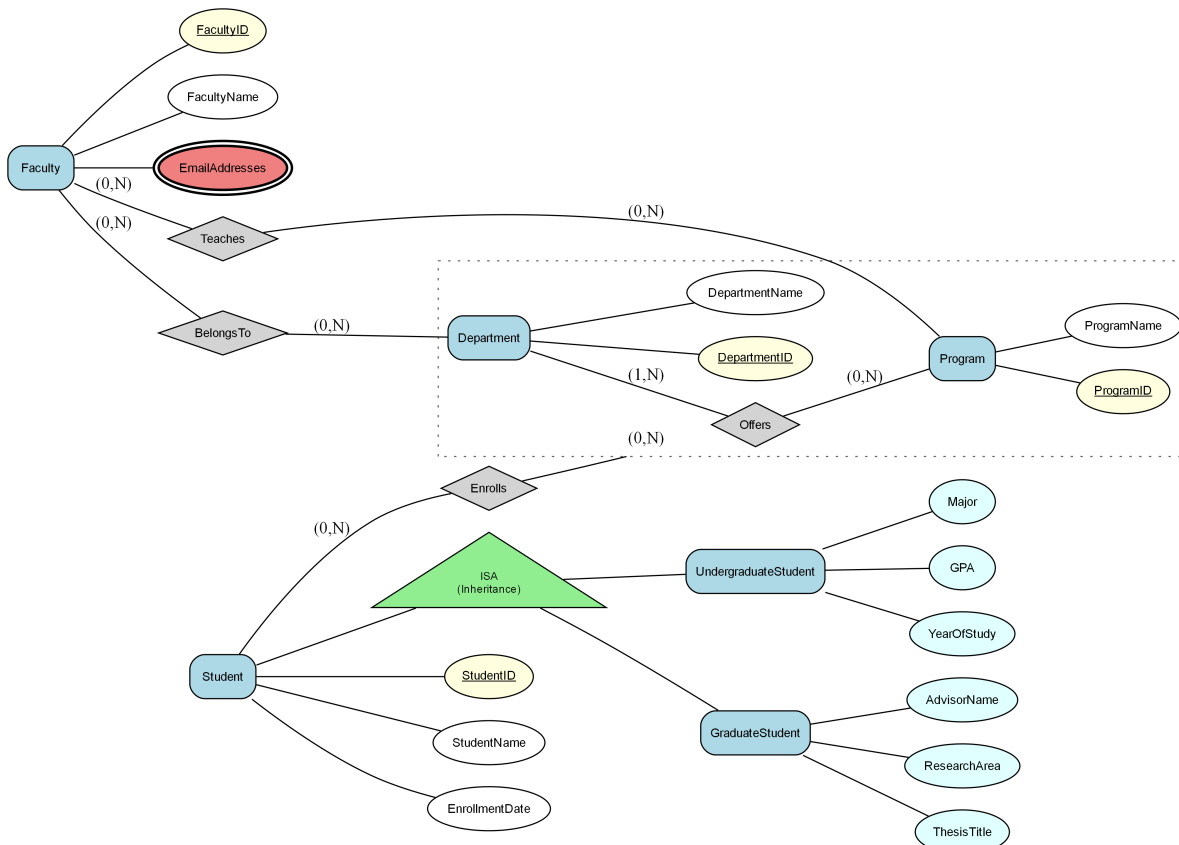
Consider the following Enhanced Entity-Relationship (EER) diagram:

Note: The diagram uses (min, max) cardinality notation where:

- (1,1) indicates one-to-one relationship (each entity participates exactly once)
- (1,N) or (1,*) indicates one-to-many relationship (one entity can relate to many)
- (0,1) indicates optional participation (zero or one)
- (0,N) or (0,*) indicates optional many participation (zero or many)

In a university system, a 'Department' entity has attributes: DepartmentID, Name, and Location. The 'Course' entity has attributes: CourseID, Title, and Credits. The 'Instructor' entity includes InstructorID, Name, and Email. The 'Student' entity has attributes: StudentID, Name, and Major. The Department offers multiple courses, and each course can be taught by multiple instructors. Students enroll in these courses. Additionally, the instructor relationship can have attributes such as OfficeHours and Specialization. There is specialization in The Instructor entity has attributes such as InstructorID, Name, Department., differentiating FullTimeInstructor with an additional Salary attribute and PartTimeInstructor with HoursWorked attribute

Figure: EER



- a) Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly. (17 marks)
- b) What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer. (3 marks)

Q2 (15 marks)

Consider a relation $R(A, B, C, D, E)$ with the following set of functional dependencies F over R :
 $F = \{ \{AB \rightarrow C, B \rightarrow D, AC \rightarrow E, D \rightarrow A, C \rightarrow B\} \}$

- a) Find all the keys in relation R using the attribute closure method. (4 marks)
- b) Is R in 3NF? Give reasons for your conclusion. (3 marks)
- c) Is R in BCNF? Give reasons for your conclusion. If R is not in BCNF, convert it to a set of BCNF relations. (8 marks)

Q3 (25 marks)

A university database is designed to manage student records, course registrations, and grades. The database schema includes tables named Students, Courses, and Enrollments, defined as follows:
Students(student_id INT PRIMARY KEY, name VARCHAR(100), major VARCHAR(50));
Courses(course_id INT PRIMARY KEY, title VARCHAR(100), credits INT); Enrollments(enrollment_id INT PRIMARY KEY, student_id INT, course_id INT, grade CHAR(2), FOREIGN KEY(student_id) REFERENCES Students(student_id), FOREIGN KEY(course_id) REFERENCES Courses(course_id)).
Use this schema to answer the following questions.

- a) In Type 2 Driver, JDBC API calls are converted to native Java API calls. Accept or refute the above statement justifying your answer. (2 marks)

- b) Code Segment:

```
java
```

String s

There are several different statements in the JDBC API to retrieve result sets based on different requirements. Which type of statements is used in the code segment shown above? Briefly explain when this type of statement will be used. (2 marks)

- c) Briefly explain the options available for transferring data between two tables within the same database. (2 marks)

- d) Briefly explain how integrity constraints are applied in SQL databases. (3 marks)

e) A university is establishing role-based access control for its database system that manages Courses, Students, and Enrollments. Sarah is the senior database administrator responsible for overall database administration. Emily is a junior database administrator responsible for managing the Enrollments database. Nathan is a database developer responsible for creating and maintaining database objects. Michael is a data entry operator responsible for entering and updating student enrollment data.

i. Write a T-SQL statement to create a login to Sarah with windows authentication. (2 marks)

ii. Provid
fixed serv

Q4 (40 marks)

Consider the following schema of a database designed for a Hospital: Patient (patientId: int, name: varchar(100), address: varchar(150), dob: date) Doctor (doctorId: int, name: varchar(100), specialty: varchar(100), phone: varchar(15)) Appointment (appointmentId: int, patientId: int, doctorId: int, appointmentDate: date, status: varchar(20)) Fee (feeld: int, appointmentId: int, amount: real, paymentStatus: varchar(20)) The 'Patient' table stores information about patients, including a unique patient ID, name, address, and date of birth. The 'Doctor' table contains details about doctors, including their unique ID, name, specialty, and phone number. The 'Appointment' table records appointment details, linking patients with doctors, including appointment date and status. The 'Fee' table keeps track of fees associated with appointments, including the amount and payment status. The 'Appointment' table manages appointment records, tracking scheduled appointments between patients and doctors with dates and times. The 'Fee' table contains relevant information for the database system.

a) Write SQL Queries to perform the following:

i. Find the names of doctors who have appointments scheduled on '2023-10-01'. (4 marks)

ii. Find
(6 marks)

b) Create a function to calculate the total fee amount for a given patient. This function should account for fee with a late penalty of 10% applied to those marked as 'overdue' payment status. (11 marks)

c) Create a trigger that automatically updates the 'amount' column in the 'Fee' table whenever a new fee record is added, or an existing fee record is updated. The 'amount' column should contain the total fee amount for each patient. Ensure the trigger adjusts the 'amount' column accurately to reflect any partial payments made by patient. The patient pays only 50% of the total fee amount as an initial payment when the payment status is marked as 'Partial'. (12 marks)